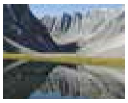
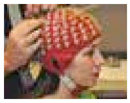


Mercury pollution in the Arctic
Mercury emissions from industrialised nations are making their way to the arctic in an alarming manner. **http://dgit.in/ArcticPoison**



Less invasive brain electrodes
Researchers have come up with a new method that allows them to insert tiny, flexible electrodes into the brain **http://dgit.in/BrnElec**

wards certain things as addiction. For instance, how many times have you heard, "I'm addicted to this song!" or "I can't do without my morning coffee, I'm addicted to it". While there is nothing wrong with either statements spoken casually, it also belittles a situation where a person might actually be addicted and refuse to acknowledge it as a problem. So first things first, beyond obvious signs, if you or anyone you know might repeatedly refer to something as addiction, sit up and take notice.

THE BIOLOGY

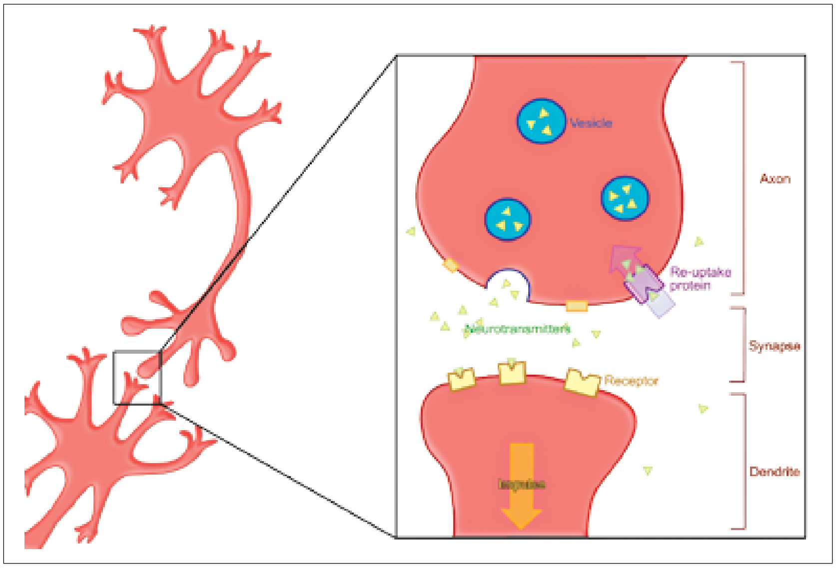
While quite a lot of physical effects that are present in drug-induced addiction are missing from behavioural addiction, they are both, at a neural level, the same. And to understand one would be to understand another up to some extent. So, what causes us to be addicted? We start off with understanding neurotransmitters.

Our body functions the way it does largely due to our brain controlling it via the nervous system. In this system, every part of our body is linked to our brain with nerves either via our spinal cord or directly. These nerves, in turn, are made up of nerve cells. But these nerve cells, or neurons, are not connected continuously – they have gaps in between them called synapses. And when a signal to or from the brain needs to cross this synapse, it is carried over by chemicals known as neurotransmitters.

Drugs can play havoc with neurotransmitters – flood the brain with excess neurotransmitters, stop the brain from making them, bind to receptors, causing the neurotransmitters to be destroyed and way more. And that is what causes addiction.

THE ROLE OF DOPAMINE

All major drugs of abuse create excess dopamine in the body that causes a feeling of pleasure. In excess, drug abuse leads to high level of dopamine in the body. But wait, if it makes you feel happy all the time, what's wrong with that? Enter homeostasis.



A simplistic diagram of how two neurons interact

Without going into too much detail, homeostasis can be explained as the tendency of the human body to maintain its state. That includes the brains and the neurotransmitters it produces as well. Once acclimated to, say, a high supply of dopamine from a drug, the brain reduces its own production capacity to maintain a normal level. But this causes low dopamine level when the drug is not being used, eventually leading the user into depression, stress and the belief that the only relief they can get is through the drug.

And this is not just limited to drug addiction. A 2009 study by the Scripps Research Institute concluded that the same molecular reactions that occur with human drug addiction also show up in compulsive overeating in obese rats. The receptor that was studied – D2 – is responsible for vulnerability to drug addiction in humans and responds to dopamine, a central neurotransmitter released in anticipation of rewarding, satiating experiences such as those involving food, sex or psychoactive drugs. When checked, it was found to be unregulated in obese rats exposed to a high fat diet, and the sensitivity to dopamine went even lower when the rats were allowed to continue their compulsive eating behaviour.

In a now classic experiment that has been repeated a number of times, researchers attached electrodes to rodents in such a way that when activated, they stimulated the reward centres of the brain. This activation can be done by a button in the enclosure, which is repeatedly done by some rodents until the reach the point of passing out due to exhaustion. This is important, since biochemical processes work in a remarkably similar way in rodents and humans too. Even

Smartphone addiction has even created a new form of muscular ailment known as text neck